

Unsigned Distance Maps on 2D Point Cloud Registration

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Abstract—2D point cloud registration is the front-end of Simultaneous Localization and Mapping (SLAM) for mobile robots with 2D lasers, where the back-end consumes both the pose estimate and its uncertainty. Iterative Closest Point (ICP), one of the most used approaches, recomputes correspondences via nearest-neighbor search at every iteration, whereas correspondence-free alternatives focus on scan-to-map alignment. This paper proposes 2D point cloud registration based on unsigned distance maps, precomputing the distance to the nearest reference point over a grid, along with its derivatives, replacing the per-iteration search with $O(1)$ lookups. Point-to-point and point-to-plane formulations are derived on $SE(2)$, solved via Gauss-Newton, and evaluated for accuracy and uncertainty consistency against ICP baselines. On a synthetic benchmark and IILABS 3D dataset, the precomputed point-to-point variant outperforms its analytical counterparts, achieving competitive odometry drift compared to point-to-plane formulations, as the precomputed gradient regularizes correspondences in the presence of noise.

Index Terms—2D laser scanner, mobile robots, point cloud registration, unsigned distance fields.

I. INTRODUCTION

2D point cloud registration estimates the rigid-body transformation that best aligns two point clouds acquired by a range sensor. This problem arises in laser odometry and Simultaneous Localization and Mapping (SLAM), where scans from a 2D laser are aligned to track the robot's pose and build a consistent representation of the environment. For ground mobile robots equipped with 2D lasers, such as industrial autonomous mobile robots, registration serves as the front-end of a SLAM pipeline, whose back-end consumes both the relative pose and its information matrix estimates. Registration errors accumulate as odometric drift, while overconfident information matrices distort the back-end optimization.

Iterative Closest Point (ICP) [1], [2] remains the most widely used registration approach, alternating between data association, which establishes point correspondences via

a nearest-neighbor search (commonly accelerated with a kd-tree [3] at $O(\log n)$ cost per point), and an error-minimization step using a point-to-point [1] or point-to-plane [2] metric. Methods such as the Normal Distributions Transform (NDT) [4] and probabilistic formulations [5], [6] avoid explicit correspondences but are typically oriented toward scan-to-map alignment. In contrast, precomputed spatial-gradient approaches, such as HectorSLAM [7] and Perfect-Match [8], [9], evaluate the registration error against occupancy grid maps or precomputed distance fields, thereby removing per-iteration association. Nonetheless, these approaches were oriented towards scan-to-map registration under specific assumptions, leaving open the design of a unified representation that removes the per-iteration nearest-neighbor search while supporting both point-to-point and point-to-plane residuals within a common least-squares framework.

This work proposes a 2D point cloud registration approach based on unsigned distance maps. A distance map encodes the Euclidean distance to the nearest reference point, decoupling data association from the optimization loop. During registration, each moving point is projected and queried in the distance field at $O(1)$ cost, replacing the nearest-neighbor search performed at every ICP iteration. We derive point-to-point and point-to-plane error formulations on the $SE(2)$ manifold and solve them with a Gauss-Newton optimizer within the `srrg2_solver` framework [10]. This work focuses on the accuracy and uncertainty of the resulting estimates, as Sobreira *et al.* [9] already showed lookup tables to make ICP 5–9× faster than kd-trees in localization, at the same accuracy.

The main contributions of this work are the following:

- a unified unsigned distance map open-source framework¹ for 2D point cloud registration, with three representation levels and dense or sparse storage, and serving as a drop-in replacement for the kd-tree [3] search of ICP [1], [2];
- point-to-point and point-to-plane residuals and their Jacobians on the $SE(2)$ manifold built on the distance field, with the analytical point-to-plane proposed formulation shown to be equivalent to classical point-to-plane ICP [2];
- an evaluation of accuracy and uncertainty consistency on a synthetic benchmark and the IILABS 3D [11] dataset, where the precomputed point-to-point variant lowers odometric drift relative to its analytical counterparts in most configurations and yields less overconfident information matrices on degenerate scenes.

¹<https://github.com/INESCTEC/ricoslam>

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II. RELATED WORK

One of the most popular approaches for point cloud registration is the ICP algorithm. Besl and McKay [1] introduced ICP for 3D shapes, iterating between (i) computing correspondences via nearest-neighbor and (ii) estimating the rigid transformation that minimizes a point-to-point distance between corresponding points. Chen and Medioni [2] proposed a point-to-plane variant exploiting surface normals on the target cloud, minimizing the squared dot product of the normal with the residual between projected source and target points. Comparing ICP variants, Rusinkiewicz and Levoy [12] found point-to-plane [2] to outperform the point-to-point [1] metric.

Unlike ICP methods that rely on explicit correspondences, NDT by Biber and Strasser [4] performs 2D registration by discretizing space into cells, each modelled as a Gaussian over its points, and minimizing an error function derived from these distributions via Gauss-Newton. Magnusson *et al.* [13] extended NDT [4] to 3D using voxels, aligning scans by evaluating point likelihoods under the other scan's distributions. Compared to point-to-point ICP [1] with kd-tree search [3], 3D-NDT [13] lowered computation cost and offered a more compact representation by avoiding explicit correspondences.

While using NDT [4], [13] representations, GICP [14] and NICP [15] retain the ICP iterative procedure. GICP [14] kept kd-tree [3] correspondence search but introduced a plane-to-plane metric, modelling local surface covariances via Principal Component Analysis (PCA) to weight residuals by geometric uncertainty. GICP proved more robust and accurate than point-to-point and point-to-plane ICP [1], [2]. NICP [15] extended GICP [14] in both association (a line-of-sight step with normal- and curvature-based rejection) and optimization, minimizing a Mahalanobis distance in a 6D point-normal space. NICP outperformed GICP [14] and NDT [13] in accuracy.

Probabilistic methods have also been proposed for 2D registration. Montesano *et al.* [5] build correspondences from a Mahalanobis-based test and maximize the alignment probability over the correspondence distribution, converging faster than ICP [1]. Censi [6] considered all possible correspondences, generating alignment hypotheses through the Generalized Hough Transform (GHT) within a probabilistic framework. GMapping [16] and Cartographer [17] align scans against the map by maximum likelihood, via gradient descent over the beam-endpoint likelihood [18] and non-linear least squares over smoothed occupancy values, respectively.

Precomputed spatial gradients enable 2D registration without explicit correspondences. HectorSLAM [7] maximizes the occupancy probability of beam endpoints over the 2D grid map, using bilinear interpolation and approximate derivatives in a Gauss-Newton update. PerfectMatch [8], [9] formulates 2D registration on distance maps for robot soccer, evaluating each point through a distance map giving the distance to the closest field marking under a robust M-estimator. This method considered only scan-to-map alignment and approximated the estimation covariance assuming orthogonal walls. Benchmarking against ICP [1] and NDT [4], Sobreira *et al.* [9]

found PerfectMatch [8] faster given the usage of precomputed gradients and more tolerant to orientation initialization errors.

Overall, most ICP methods require two steps per iteration. Probabilistic approaches typically solve a maximum a posteriori problem and need a map estimate, which is incompatible with scan-to-scan registration [14]. In contrast, distance maps provide precomputed lookup tables that evaluate each source point's registration error without explicit association, supporting both scan-to-scan and scan-to-map alignment.

III. UNSIGNED DISTANCE MAPS

Let $\mathcal{P} = \{\mathbf{p}_1, \dots, \mathbf{p}_n\} \subset \mathbb{R}^2$ denote a reference target 2D point cloud acquired by a planar laser scanner and expressed in the sensor frame S . The distance map $\mathcal{D} : \mathbb{R}^2 \rightarrow \mathbb{R}_0^+$ maps a query point $\mathbf{y} \in \mathbb{R}^2$ represented in the sensor frame S to its Euclidean distance to the nearest point in $\mathcal{P}$, as in (1), where $\mathbf{p}^*(\mathbf{y}) = \arg\min_{\mathbf{p}_i \in \mathcal{P}} \|\mathbf{y} - \mathbf{p}_i\|_2$ is the nearest neighbor.

$$\mathcal{D}(\mathbf{y}) = \min_{\mathbf{p}_i \in \mathcal{P}} \|\mathbf{y} - \mathbf{p}_i\|_2 = \|\mathbf{y} - \mathbf{p}^*(\mathbf{y})\|_2 \quad (1)$$

Since $\mathcal{D}(\mathbf{y}) \geq 0$ for all $\mathbf{y}$, it is an *unsigned distance function*, as opposed to signed distance functions used in volumetric reconstruction. Unsigned distance fields are continuous everywhere but not everywhere differentiable [19].

We precompute $\mathcal{D}$ once over a discrete grid of cell resolution δ (m/px). Each cell stores information about the nearest reference point, solving data association implicitly: a moving point transformed by the current pose estimate is looked up in its cell, which returns the distance and its derivatives in $O(1)$ without requiring further reference to $\mathcal{P}$.

A. Spatial derivatives

Wherever the nearest neighbor is unique, $\mathcal{D}$ is smooth and its gradient is the unit vector pointing from $\mathbf{p}^*(\mathbf{y})$ towards $\mathbf{y}$, as in (2). The unit-norm property $\|\nabla \mathcal{D}\|_2 = 1$ holds almost everywhere (Eikonal property of Euclidean distance fields [19]) except at points equidistant from two or more reference points, where the gradient is undefined.

$$\nabla \mathcal{D}(\mathbf{y}) = \frac{\mathbf{y} - \mathbf{p}^*(\mathbf{y})}{\|\mathbf{y} - \mathbf{p}^*(\mathbf{y})\|_2}, \quad \|\nabla \mathcal{D}(\mathbf{y})\|_2 = 1 \quad (2)$$

In the precomputed discrete grid, we estimate the gradient $\nabla \mathcal{D}$ and the Hessian $\mathbf{H}_{\mathcal{D}} \in \mathbb{R}^{2 \times 2}$ by central finite differences over neighboring cells, with one-sided differences at grid borders. We compute the Hessian as the centered difference of the normalized gradients $\hat{\nabla} \mathcal{D}$. Unlike the analytical expression in (3), this approximation does not guarantee symmetry, motivating our representation level that stores all four entries.

$$\mathbf{H}_{\mathcal{D}}(\mathbf{y}) = \begin{bmatrix} \frac{(y_2 - p_y^*(\mathbf{y}))^2}{\|\mathbf{y} - \mathbf{p}^*(\mathbf{y})\|_2^3} & -\frac{(y_1 - p_x^*(\mathbf{y}))(y_2 - p_y^*(\mathbf{y}))}{\|\mathbf{y} - \mathbf{p}^*(\mathbf{y})\|_2^3} \\ -\frac{(y_1 - p_x^*(\mathbf{y}))(y_2 - p_y^*(\mathbf{y}))}{\|\mathbf{y} - \mathbf{p}^*(\mathbf{y})\|_2^3} & \frac{(y_1 - p_x^*(\mathbf{y}))^2}{\|\mathbf{y} - \mathbf{p}^*(\mathbf{y})\|_2^3} \end{bmatrix} \quad (3)$$

B. Representation levels

Each cell may store different amounts of precomputed information. We consider for this work three representation levels $\mathcal{M}_l$ of increasing information and memory cost, which determine the quantities available at $O(1)$ query cost:

TABLE I
COMPARISON OF THE NANO, STANDARD, AND SPARSE
DISTANCE MAP VARIANTS

Variant	Storage	Values/cell	Query	$d_{\max}$?
<i>Nano</i> ($\mathcal{M}_{0,\text{dense}}$)	Dense array	1 (int)	$O(1)$	optional
<i>Standard</i> ($\mathcal{M}_{1 2,\text{dense}}$)	Dense array	3–8*	$O(1)$	optional
<i>Sparse</i> ($\mathcal{M}_{1 2,\text{sparse}}$)	Hash map	3–8*	$O(1)^\dagger$	advisable

* Depends on the representation level: 3 values for $\mathcal{M}_1$, 7 for $\mathcal{M}_2$, +1 if the optional nearest-neighbor index $k^*(\mathbf{y})$ is stored.

† average $O(1)$; worst-case $O(N_d)$ under hash collision, mitigated by the post-propagation rehash, where $N_d \leq N_G$ represent the number of populated cells within $d_{\max}$ and N_G the total number of cells.

- **Level 0** ($\mathcal{M}_0$): each cell stores only the integer index $k^*(\mathbf{y})$ of the nearest reference point in $\mathcal{P}$. The distance, gradient, and Hessian are recovered on demand from their analytical expressions (see (1), (2) and (3)). $\mathcal{M}_0$ is the most memory-efficient layout and serves as an $O(1)$ drop-in replacement for the kd-tree [3] search of ICP, reducing the per-query cost from $O(\log n)$ to $O(1)$;
- **Level 1** ($\mathcal{M}_1$): each cell stores the scalar distance $\mathcal{D}$ together with the two components of the gradient $\nabla \mathcal{D}$ (optionally also k^*). The Hessian, if needed, is recovered at query time through centered difference approximation;
- **Level 2** ($\mathcal{M}_2$): each cell additionally stores all entries of the finite-difference Hessian $\mathbf{H}_{\mathcal{D}}$, ensuring the second-order information is available at $O(1)$.

C. Data structures

Independently of the representation level, we build the three distance map representations considered in this work by a best-first wavefront propagation over an 8-connected neighborhood, analogous to Dijkstra’s shortest-path algorithm [20]: the cells occupied by reference points are seeded at zero distance and the nearest-neighbor index is propagated outward, either over the whole map or until a truncation distance $d_{\max}$ is reached. Then, post-processing passes compute the gradient and Hessian by finite differences.

We implement the distance map representations with three data structures, summarized in Table I: *Nano* ($\mathcal{M}_{0,\text{dense}}$), a dense integer-index array with the lowest memory footprint; *Standard* ($\mathcal{M}_{1|2,\text{dense}}$), a dense array of full cells; and *Sparse* ($\mathcal{M}_{1|2,\text{sparse}}$), a hash table (`std::unordered_map` in C++) that stores only the cells within the truncation band $\mathcal{D} \leq d_{\max}$, analogous to the truncated signed distance fields used in volumetric reconstruction [21]. All three data structures provide $O(1)$ lookup, with the sparse variant achieving it on average and mitigating hash-collision worst cases through a post-propagation rehash.

IV. REGISTRATION METHODOLOGY

This section describes the proposed 2D point cloud registration. We introduce two constraint factors on $SE(2)$ built on top of the unsigned distance map $\mathcal{D}$ of Section III: a point-to-point residual (Section IV-B) and a point-to-plane residual (Section IV-C). Both replace the per-iteration nearest-neighbor search of ICP with $O(1)$ grid lookups of $\mathcal{D}$ and its derivatives.

We implement these constraint factors on the `srrg2_solver` framework [10] with the manifold Gauss-Newton optimizer.

A. Problem formulation

Let $\mathbf{z}_i \in \mathbb{R}^2$, $i = 1, \dots, M$, denote the points of the moving (current) cloud expressed in the sensor frame S . The goal of registration is to find the robot pose $\mathbf{x} \in SE(2)$ that best aligns the moving cloud to the reference one. The pose is a homogeneous transformation $\mathbf{x} = [\mathbf{R} \mid \mathbf{t}]$, with $\mathbf{R}(\theta) \in SO(2)$ and $\mathbf{t} \in \mathbb{R}^2$. Let $\mathbf{T}_S^R \in SE(2)$ be the known sensor-to-robot transformation, with inverse $\mathbf{T}_R^S = [\mathbf{R}_R^S \mid \mathbf{t}_R^S]$. A moving point $\mathbf{z}_i$ projected by the current estimate and expressed in the reference sensor frame is represented by $\tilde{\mathbf{z}}_i$, as in (4).

$$\tilde{\mathbf{z}}_i = \mathbf{T}_R^S \mathbf{x} \mathbf{T}_S^R \mathbf{z}_i, \quad \mathbf{z}_{R,i} = \mathbf{T}_S^R \mathbf{z}_i. \quad (4)$$

The problem is formulated as a non-linear least-squares problem and solved with a Gauss-Newton optimizer on the $SE(2)$ manifold. Because the optimization variable lives on a manifold, increments cannot be applied by Euclidean addition. A perturbation $\Delta \mathbf{x} = (\Delta x, \Delta y, \Delta \theta)^T \in \mathbb{R}^3$ is defined in the tangent space and applied through a right-sided update $\mathbf{x}_{k+1} = \mathbf{x}_k \oplus \Delta \mathbf{x} = \mathbf{x}_k \cdot \exp(\Delta \mathbf{x})$, with $\exp(\Delta \mathbf{x}) = [\mathbf{R}(\Delta \theta) \mid (\Delta x, \Delta y)^T]$. For the details of the manifold least-squares machinery (the Gauss-Newton normal equations, robust kernels, and information-matrix recovery for uncertainty estimation), we follow its formulation and the `srrg2_solver` implementation of Grisetti *et al.* [10]. We use the 2D skew operator $[\mathbf{v}] = (-v_y, v_x)^T$, which encodes the rotational sensitivity of a vector and populates the column associated with $\Delta \theta$ in the Jacobians derived below.

B. Point-to-point error formulation

First, we propose a point-to-point residual as the scalar returned by the distance map $\mathcal{D}$ at the projected moving point $\tilde{\mathbf{z}}_i$, as in (5). This residual measures the Euclidean distance from $\tilde{\mathbf{z}}_i$ to its nearest reference point, as encoded by the precomputed field $\mathcal{D}$.

$$e_i^{\text{p2p}}(\mathbf{x}) = \mathcal{D}(\tilde{\mathbf{z}}_i) = \mathcal{D}(\mathbf{T}_R^S \cdot \mathbf{x} \cdot \mathbf{T}_S^R \mathbf{z}_i) \in \mathbb{R} \quad (5)$$

Applying the right-sided perturbation and differentiating w.r.t. $\Delta \mathbf{x}$ at $\Delta \mathbf{x} = \mathbf{0}$ through the chain rule yields the 1×3 Jacobian of (6). The first factor is the distance-map gradient at the projected point. The second one, $\mathbf{J}_i^{\text{p2p,ICP}}$, coincides with the Jacobian of the classical point-to-point ICP error [1]. The gradient $\nabla \mathcal{D}(\tilde{\mathbf{z}}_i)$ is retrieved at $O(1)$ from the $\mathcal{M}_1$ (or $\mathcal{M}_2$) representation, or evaluated analytically at $\mathcal{M}_0$.

$$\mathbf{J}_i^{\text{p2p}} = \nabla \mathcal{D}(\tilde{\mathbf{z}}_i)^T \cdot \underbrace{\mathbf{R}_R^S \mathbf{R}(\theta) \cdot (\mathbf{I}_{2 \times 2} \mid [\mathbf{z}_{R,i}])}_{\mathbf{J}_i^{\text{p2p,ICP}}} \in \mathbb{R}^{1 \times 3} \quad (6)$$

C. Point-to-plane error formulation

Our point-to-plane variant incorporates surface-normal information to improve convergence in environments dominated by planar structures, analogously to Chen and Medioni [2]. Normals $\mathbf{n}_{z_i}$ are estimated on the moving cloud (Section IV-D). Let $\hat{\nabla} \mathcal{D}(\tilde{\mathbf{z}}_i)$ be the normalized gradient. The

proposed residual projects the rotated moving normal onto the vector from the moving point to its nearest reference point, encoded by $\hat{\nabla}\mathcal{D} \cdot \mathcal{D}$, as in (7).

$$e_i^{\text{p2pl}}(\mathbf{x}) = (\mathbf{R}(\theta) \mathbf{n}_{z_i})^T \hat{\nabla}\mathcal{D}(\tilde{\mathbf{z}}_i) \mathcal{D}(\tilde{\mathbf{z}}_i) \quad (7)$$

Differentiating (7) w.r.t. $\Delta\mathbf{x}$ gives the 1×3 Jacobian of (8). The first term accounts for the rotation of the normal under $\Delta\theta$. The second and third represent the change in gradient direction (via the Hessian $\mathbf{H}_{\mathcal{D}}$) and in distance value, respectively, both sharing the factor $\mathbf{J}_i^{\text{p2p,ICP}}$.

$$\begin{aligned} \mathbf{J}_i^{\text{p2pl}} &= \mathbf{n}_{z_i}^T \left(\mathbf{0}_{2 \times 2} \mid [-\mathbf{R}(\theta)^T \hat{\nabla}\mathcal{D}(\tilde{\mathbf{z}}_i) \mathcal{D}(\tilde{\mathbf{z}}_i)] \right) \\ &+ (\mathbf{R}(\theta) \mathbf{n}_{z_i})^T \mathbf{H}_{\mathcal{D}}(\tilde{\mathbf{z}}_i) \mathcal{D}(\tilde{\mathbf{z}}_i) \mathbf{J}_i^{\text{p2p,ICP}} \\ &+ (\mathbf{R}(\theta) \mathbf{n}_{z_i})^T \hat{\nabla}\mathcal{D}(\tilde{\mathbf{z}}_i) \nabla\mathcal{D}(\tilde{\mathbf{z}}_i)^T \mathbf{J}_i^{\text{p2p,ICP}} \end{aligned} \quad (8)$$

When the $\mathcal{M}_0$ representation is used, the gradient and distance are evaluated analytically and the residual of (7) reduces to the dot product of the rotated moving normal with the displacement $(\tilde{\mathbf{z}}_i - \mathbf{p}^*)$, which is exactly the classical point-to-plane ICP residual [2]. Thus, the two formulations are analytically identical at $\mathcal{M}_0$, a property confirmed numerically in Section V. The second and third terms require the Hessian and gradient, both retrieved at $O(1)$ from the $\mathcal{M}_2$ representation.

D. Surface-normal estimation

Normals are estimated once per scan, before optimization. Exploiting the angular ordering of the 2D laser scan, a spatially coherent neighborhood $\mathcal{N}_r(\mathbf{z}_i)$ of radius r is collected by traversing the ordered scan in both directions from i (wrapping at the scan boundaries) until the Euclidean distance to $\mathbf{z}_i$ exceeds r , avoiding a kd-tree search [3]. If the neighborhood has at least $N_{\min} \geq 3$ points, the 2×2 covariance $\mathbf{C}_i$ is computed and the normal $\mathbf{n}_{z_i}$ is taken as the eigenvector of least variance via PCA. The sign ambiguity is resolved by orienting the normal toward the sensor origin ($\mathbf{n}_{z_i}^T \mathbf{z}_i \leq 0$), which reflects that obstacle-boundary normals consistently face the scanner. Points with insufficient neighbors are marked invalid for the point-to-plane term.

V. EXPERIMENTAL RESULTS

We evaluate the proposed distance map registration framework on two experiments. A synthetic benchmark (Section V-A) measures pairwise registration accuracy and uncertainty estimation against known ground-truth poses under controlled conditions, whereas a real-world laser-odometry evaluation (Section V-B) assesses odometric drift over extended trajectories. Both experiments compare the proposed distance map point-to-point ($\mathcal{D}_{\text{p2p}}$, Section IV-B) and point-to-plane ($\mathcal{D}_{\text{p2pl}}$, Section IV-C) formulations against the standard point-to-point [1] and point-to-plane [2] ICP error metrics ($\text{ICP}_{\mathcal{D}, \text{p2p}}$ and $\text{ICP}_{\mathcal{D}, \text{p2pl}}$, respectively), whose data association uses the same $O(1)$ $\mathcal{M}_0$ lookup, so that accuracy differences isolate the error formulation. All formulations are solved with Gauss-Newton on the $SE(2)$ manifold within `srrg2_solver` [10], ensuring consistent conditions across methods. We evaluate $\mathcal{D}_{\text{p2p}}$ at representation levels $\mathcal{M}_0$ and $\mathcal{M}_1$, $\mathcal{D}_{\text{p2pl}}$ at $\mathcal{M}_0$ and $\mathcal{M}_2$, and both ICP baselines at $\mathcal{M}_0$.

A. Synthetic benchmark

a) *Experimental setup*: Synthetic 2D scans are generated with a laser-scanner range model parametrized based on the Hokuyo UST-10LX sensor (30 m maximum range, 0.25° angular resolution, and 0.03 m repeatability used as the range-noise standard deviation) and with a 360° field-of-view. Three planar scenes are considered: a *circle* (10 m radius observed from its interior, rotationally symmetric, and thus, rotationally degenerate); a *square* (20 m side, constraining all three degrees of freedom of the $SE(2)$ pose up to symmetry); and a *corridor* (two parallel 100 m walls 10 m apart, where translation along the corridor axis is unobservable because the 30 m range does not reach the endpoints). Each scene is evaluated over $N = 100$ independent trials with the true pose $\mathbf{x}^* \in SE(2)$ drawn uniformly (translations up to ± 1.5 m, reduced for the corridor scene; rotations up to $\pm 20^\circ$). Zero-mean Gaussian noise with $\sigma \in \{0, 0.03 \text{ m}\}$ is added independently to both scans, modelling the noise-free and Hokuyo UST-10LX cases.

All methods adopt Gauss-Newton on $SE(2)$ (up to 50 iterations, step-norm termination threshold 10^{-5}) and a $d_{\max} = 5$ m truncation distance. Surface normals for the point-to-plane formulations are estimated via PCA over a 0.15 m neighborhood (minimum of 3 points). Results are reported with and without point cloud voxelization at the cell resolution δ and with and without bi-linear sub-pixel interpolation (flags V and S in the tables; S applies only to $\mathcal{M}_1$ and $\mathcal{M}_2$). Table II and Table III report the results of the synthetic benchmark for the $\delta = 0.03$ m and $\delta = 0.05$ m resolutions.

b) *Evaluation metrics*: Given an estimate $\mathbf{x}$ and its ground-truth $\mathbf{x}^*$, the target-frame residual transformation $\mathbf{x} \cdot \mathbf{x}^{*-1} = [\mathbf{R}_\epsilon \mid \mathbf{t}_\epsilon]$ yields the pose error $\epsilon(\mathbf{x})$ of (9) [22], from which we derive the scalar metrics in (10).

$$\mathbf{x} \cdot \mathbf{x}^{*-1} = [\mathbf{R}_\epsilon \mid \mathbf{t}_\epsilon], \quad \epsilon(\mathbf{x}) = (\mathbf{t}_\epsilon^T, \theta_\epsilon)^T \in \mathbb{R}^3 \quad (9)$$

$$\epsilon_t = \|\mathbf{t}_\epsilon\|_2, \quad \epsilon_y = |t_{\epsilon,y}|, \quad \epsilon_\theta = |\text{atan2}(r_{\epsilon,2,1}, r_{\epsilon,1,1})| \quad (10)$$

The translation error ϵ_t is reported for the circle and square scenes. For the corridor, only the observable perpendicular component ϵ_y is used, since the corridor axis is unobservable. The rotation error ϵ_θ is omitted for the circle, whose rotational DoF is degenerate. The internal consistency of each solver's uncertainty estimate is assessed by the Hessian-weighted residual norm of (11), where $\mathbf{\Omega} \in \mathbb{R}^{3 \times 3}$ is the information matrix returned by the Gauss-Newton solver [10], [22].

$$\chi_\epsilon^2 = \epsilon(\mathbf{x})^T \mathbf{\Omega} \epsilon(\mathbf{x}) \quad (11)$$

A value $\chi_\epsilon^2 = 0$ denotes a perfect alignment, while large values indicate poor accuracy or an overconfident $\mathbf{\Omega}$. The metric inflates whenever the alignment fails along a degenerate direction, making it a relative cross-formulation assessment of uncertainty calibration. The errors ϵ_t , ϵ_y , and ϵ_θ are aggregated as the Root-Mean-Square Error (RMSE) over the $N = 100$ trials, whereas χ_ϵ^2 is reported as the mean.

TABLE II
SYNTHETIC BENCHMARK RESULTS FOR POINT CLOUD ALIGNMENT USING PROPOSED DISTANCE MAP-BASED ERROR FORMULATIONS
AND DISTANCE MAP-GUIDED ICP BASELINES [1], [2] (CELL RESOLUTION $\delta = 0.03$ m)

Meth.	$\mathcal{M}_l$	V?	S?	Circle				Square						Corridor					
				$\sigma = 0.00$ m		$\sigma = 0.03$ m		$\sigma = 0.00$ m			$\sigma = 0.03$ m			$\sigma = 0.00$ m			$\sigma = 0.03$ m		
				ε_t (m)	χ_ε^2	ε_t (m)	χ_ε^2	ε_t (m)	ε_θ (°)	χ_ε^2	ε_t (m)	ε_θ (°)	χ_ε^2	ε_y (m)	ε_θ (°)	χ_ε^2	ε_y (m)	ε_θ (°)	χ_ε^2
$\mathcal{D}_{p2p}$	$\mathcal{M}_0$	$\times$	–	0.001	6234.68	0.003	3100.86	0.001	0.19	0.94	0.003	0.08	0.19	0.007	0.11	540.06	0.003	0.12	453.71
		$\checkmark$	–	0.009	4827.02	0.010	3123.95	0.012	0.01	0.10	0.006	0.02	0.04	0.009	0.10	462.70	0.003	0.11	461.08
	$\mathcal{M}_1$	$\times$	$\times$	0.018	586.23	0.003	934.92	0.015	0.02	0.07	0.003	0.01	0.00	0.008	0.21	88.20	0.005	0.20	158.98
		$\times$	$\checkmark$	0.029	257.09	0.021	536.29	0.051	0.02	0.32	0.021	0.01	0.12	0.011	0.22	70.90	0.023	0.21	113.04
		$\checkmark$	$\times$	0.018	584.83	0.003	937.83	0.015	0.02	0.07	0.003	0.01	0.00	0.008	0.20	98.33	0.005	0.20	158.77
		$\checkmark$	$\checkmark$	0.029	257.08	0.021	536.49	0.051	0.02	0.32	0.021	0.01	0.12	0.011	0.22	75.83	0.024	0.20	112.46
$\mathcal{D}_{p2pl}$	$\mathcal{M}_0$	$\times$	–	0.000	52.86	0.003	189.38	0.000	0.00	0.00	0.002	0.02	0.01	0.000	0.00	5.72	0.001	0.02	13.83
		$\checkmark$	–	0.000	54.10	0.004	190.31	0.000	0.00	0.00	0.002	0.01	0.01	0.000	0.00	5.23	0.002	0.01	12.94
	$\mathcal{M}_2$	$\times$	$\times$	0.009	463.03	0.003	1716.85	0.004	0.03	0.01	0.003	0.02	0.01	0.009	0.01	4.93	0.001	0.01	129.01
		$\times$	$\checkmark$	0.023	264.87	0.021	1120.07	0.023	0.08	0.33	0.021	0.01	0.30	0.012	0.01	8.66	0.015	0.01	83.51
		$\checkmark$	$\times$	0.008	462.02	0.003	1716.63	0.004	0.03	0.01	0.003	0.02	0.01	0.009	0.02	5.09	0.001	0.01	126.92
		$\checkmark$	$\checkmark$	0.023	265.03	0.021	1119.63	0.024	0.07	0.36	0.021	0.01	0.30	0.012	0.01	11.40	0.015	0.01	82.42
ICP $_{\mathcal{D}, p2p}$ [1]	$\mathcal{M}_0$	$\times$	–	0.000	6432.74	0.003	6479.07	0.001	0.19	2.07	0.003	0.08	0.39	0.000	0.01	834.29	0.001	0.02	854.64
		$\checkmark$	–	0.009	6578.50	0.010	6579.53	0.013	0.01	0.25	0.006	0.02	0.08	0.000	0.01	775.70	0.001	0.01	817.46
ICP $_{\mathcal{D}, p2pl}$ [2]	$\mathcal{M}_0$	$\times$	–	0.000	52.86	0.003	189.38	0.000	0.00	0.00	0.002	0.02	0.01	0.000	0.00	5.72	0.001	0.02	13.83
		$\checkmark$	–	0.000	54.10	0.004	190.31	0.000	0.00	0.00	0.002	0.01	0.01	0.000	0.00	5.23	0.002	0.01	12.94

Legend: distance map representation level ($\mathcal{M}_l$); point cloud voxelization (V); sub-pixelization (S) (bi-linear interpolation); enabled ($\checkmark$), disabled ($\times$), not applicable (–); translation error ε_t (m); rotation error ε_θ (°); pose-error consistency metric $\chi_\varepsilon^2 = \varepsilon^T \Omega \varepsilon$ (dimensionless).

TABLE III
SYNTHETIC BENCHMARK RESULTS FOR POINT CLOUD ALIGNMENT USING PROPOSED DISTANCE MAP-BASED ERROR FORMULATIONS
AND DISTANCE MAP-GUIDED ICP BASELINES [1], [2] (CELL RESOLUTION $\delta = 0.05$ m)

Meth.	$\mathcal{M}_l$	V?	S?	Circle				Square						Corridor					
				$\sigma = 0.00$ m		$\sigma = 0.03$ m		$\sigma = 0.00$ m			$\sigma = 0.03$ m			$\sigma = 0.00$ m			$\sigma = 0.03$ m		
				ε_t (m)	χ_ε^2	ε_t (m)	χ_ε^2	ε_t (m)	ε_θ (°)	χ_ε^2	ε_t (m)	ε_θ (°)	χ_ε^2	ε_y (m)	ε_θ (°)	χ_ε^2	ε_y (m)	ε_θ (°)	χ_ε^2
$\mathcal{D}_{p2p}$	$\mathcal{M}_0$	$\times$	–	0.001	6198.44	0.003	3913.39	0.001	0.10	0.35	0.003	0.29	2.15	0.018	0.07	454.93	0.003	0.14	492.12
		$\checkmark$	–	0.004	4953.51	0.026	3200.64	0.005	0.01	0.02	0.019	0.03	0.29	0.013	0.11	379.95	0.004	0.13	522.25
	$\mathcal{M}_1$	$\times$	$\times$	0.022	480.88	0.003	703.70	0.027	0.05	0.03	0.004	0.02	0.01	0.019	0.22	69.63	0.008	0.22	110.43
		$\times$	$\checkmark$	0.044	186.61	0.039	306.68	0.028	0.17	0.16	0.047	0.01	0.48	0.009	0.24	44.35	0.036	0.23	71.13
		$\checkmark$	$\times$	0.022	422.14	0.003	688.86	0.027	0.05	0.03	0.004	0.01	0.01	0.026	0.21	76.78	0.011	0.22	106.45
		$\checkmark$	$\checkmark$	0.044	162.18	0.040	304.36	0.027	0.16	0.16	0.048	0.02	0.50	0.021	0.24	45.47	0.039	0.21	69.34
$\mathcal{D}_{p2pl}$	$\mathcal{M}_0$	$\times$	–	0.000	51.30	0.003	183.62	0.000	0.00	0.00	0.002	0.02	0.01	0.000	0.00	5.72	0.001	0.02	13.80
		$\checkmark$	–	0.000	46.18	0.007	179.93	0.000	0.00	0.00	0.004	0.01	0.01	0.000	0.00	4.08	0.003	0.01	12.01
	$\mathcal{M}_2$	$\times$	$\times$	0.017	261.79	0.003	846.01	0.030	0.06	0.08	0.002	0.01	0.01	0.016	0.05	20.86	0.001	0.02	56.29
		$\times$	$\checkmark$	0.038	120.63	0.036	445.88	0.006	0.12	0.10	0.036	0.01	0.58	0.006	0.09	15.29	0.026	0.04	33.44
		$\checkmark$	$\times$	0.017	235.66	0.003	824.85	0.030	0.06	0.09	0.002	0.01	0.01	0.018	0.06	17.17	0.001	0.02	53.47
		$\checkmark$	$\checkmark$	0.037	106.47	0.036	441.65	0.007	0.12	0.10	0.035	0.01	0.57	0.007	0.10	12.93	0.025	0.05	31.53
ICP $_{\mathcal{D}, p2p}$ [1]	$\mathcal{M}_0$	$\times$	–	0.000	6431.42	0.003	6541.08	0.001	0.10	0.61	0.003	0.29	4.60	0.000	0.03	828.40	0.001	0.04	844.22
		$\checkmark$	–	0.005	5754.85	0.026	6248.71	0.005	0.01	0.04	0.020	0.03	0.58	0.000	0.01	635.54	0.002	0.01	766.20
ICP $_{\mathcal{D}, p2pl}$ [2]	$\mathcal{M}_0$	$\times$	–	0.000	51.30	0.003	183.62	0.000	0.00	0.00	0.002	0.02	0.01	0.000	0.00	5.72	0.001	0.02	13.80
		$\checkmark$	–	0.000	46.18	0.007	179.93	0.000	0.00	0.00	0.004	0.01	0.01	0.000	0.00	4.08	0.003	0.01	12.01

Legend: distance map representation level ($\mathcal{M}_l$); point cloud voxelization (V); sub-pixelization (S) (bi-linear interpolation); enabled ($\checkmark$), disabled ($\times$), not applicable (–); translation error ε_t (m); rotation error ε_θ (°); pose-error consistency metric $\chi_\varepsilon^2 = \varepsilon^T \Omega \varepsilon$ (dimensionless).

c) *Registration accuracy*: The analytical point-to-plane formulations ($\mathcal{D}_{p2pl}(\mathcal{M}_0)$ and ICP $_{\mathcal{D}, p2pl}$) achieve lower or equal ε_t and ε_θ than their point-to-point counterparts in all but one case, confirming the conclusion of Rusinkiewicz and Levoy [12] for ICP variants and extending it to distance map-based formulations. This behavior reflects the sensitivity of point-to-point residuals to the inter-scan sampling discrepancy inherent to the laser-scanner model when two scans sample the surfaces at different poses even without noise. In contrast, point-to-plane residuals measure the orthogonal point-to-surface distance. Thus, point-to-plane variants seem more invariant to that sampling discrepancy.

Moreover, $\mathcal{D}_{p2pl}(\mathcal{M}_0)$ and ICP $_{\mathcal{D}, p2pl}$ produce identical results in the synthetic benchmark, confirming the analyti-

cal equivalence derived in Section IV-C. Regarding the distance map parametrization, $\mathcal{D}_{p2p}(\mathcal{M}_1)$ accuracy degrades as δ grows. Enabling bi-linear sub-pixel interpolation does not improve, and often worsens, accuracy: at $\delta = 0.05$ m with noise and no voxelization, ε_t rises from 0.003 m, 0.004 m, and 0.008 m to 0.039 m, 0.047 m, and 0.036 m for the circle, square, and corridor scenes, respectively. This result is a consequence of initializing hit cells on the distance map computation with zero, biasing the interpolated field, even though it facilitates the precomputation procedure. Voxelization has a negligible effect under these controlled conditions, but remains recommended for 2D lasers, whose radial sampling concentrates measurements near the sensor and would otherwise bias the optimizer toward close-range observations.

The precomputed levels trade a small accuracy loss relative to the analytical $\mathcal{M}_0$ evaluation, which shares the same $O(1)$ association but computes distances and derivatives analytically. In particular, $\mathcal{D}_{p2pl}(\mathcal{M}_2)$ retains the point-to-plane advantage over the analytical point-to-point formulations under noise, yet does not match the analytical $\mathcal{D}_{p2pl}(\mathcal{M}_0)$ and $\text{ICP}_{\mathcal{D}, p2pl}$ (e.g., on the noise-free square at $\delta = 0.03$ m, its $\varepsilon_\theta = 0.03^\circ$ stays lower than 0.94° and 2.07° of the analytical point-to-point formulations, but above the 0.00° of the analytical point-to-plane). Two effects contribute to this gap. First, the grid quantization and the finite-difference gradient introduce a systematic bias in the residual direction that scales with δ . Second, the point-to-plane term at $\mathcal{M}_2$ relies on the second-order derivative, obtained by finite-differencing the already finite-difference gradient. This successive differentiation seems more noise-sensitive than the first-order gradient of $\mathcal{M}_1$, preventing $\mathcal{M}_2$ from matching the analytical point-to-plane accuracy. Thus, the analytical $\mathcal{M}_0$ point-to-plane remains the most accurate configuration in these controlled conditions, while the precomputed levels offer competitive accuracy performance.

d) Uncertainty estimation: The consistency metric χ_ϵ^2 exposes differences in the estimated information matrices. The $\mathcal{D}_{p2pl}(\mathcal{M}_0)$ and $\text{ICP}_{\mathcal{D}, p2pl}$ point-to-plane formulations yield the best-calibrated estimates across all conditions, reaching $\chi_\epsilon^2 \approx 0$ on the noise-free square scene. In contrast, the analytical point-to-point formulations are overconfident on degenerate scenes: on the circle, $\text{ICP}_{\mathcal{D}, p2p}$ yields χ_ϵ^2 up to $\approx 6.6 \times 10^3$, one to two orders of magnitude above the point-to-plane counterpart (≈ 46 – 190), as its 2×3 Jacobian accumulates a larger information matrix than the scalar 1×3 distance map residual. The precomputed $\mathcal{D}_{p2p}(\mathcal{M}_1)$ variant yields systematically lower χ_ϵ^2 than both $\mathcal{D}_{p2p}(\mathcal{M}_0)$ and $\text{ICP}_{\mathcal{D}, p2p}$ on the circle and corridor scenes (e.g., circle, noise-free, $\delta = 0.03$ m: ≈ 257 – 586 against ≈ 4.8 – 6.6×10^3), suggesting that the finite-difference gradient and the cell-resolution regularize the information matrix estimation. Conversely, $\mathcal{D}_{p2pl}(\mathcal{M}_2)$ seems more noise-sensitive than its $\mathcal{M}_0$ counterpart (e.g., on the noisy circle at $\delta = 0.03$ m, $\chi_\epsilon^2 \approx 1120$ – 1717), consistent with the compounded second-order finite differences noted in the accuracy results. Hence, the analytical $\mathcal{M}_0$ point-to-plane formulations are preferable for uncertainty-aware applications.

B. Real-world laser odometry evaluation

a) Experimental setup: We deploy the four formulations ($\mathcal{D}_{p2p}$, $\mathcal{D}_{p2pl}$, $\text{ICP}_{\mathcal{D}, p2p}$ [1], and $\text{ICP}_{\mathcal{D}, p2pl}$ [2]) as a 2D laser-odometry tracker on the IILABS 3D [11] dataset, using only the Hokuyo UST-10LX scans, with ground-truth from an OptiTrack motion-capture system. Four planar sequences are considered: *Nav A Diff* (275 m, differential drive motion), *Nav A Omni* (112 m, holonomic motion), *Loop* (232 m, where the robot exits and re-enters the Nav A space), and *Slippage* (39 m, a straight path with induced wheel slippage). The tracker uses $\delta = 0.03$ m, $d_{\max} = 0.50$ m, and a maximum sensor range $r_{\max} = 30$ m, with bi-linear interpolation disabled and voxelization enabled following the synthetic findings. Outliers are handled by a two-stage Cauchy robustifier (5 iterations

coarse stage with $c^2 = 0.250$, followed by a fine stage of up to 15 iterations with $c^2 = 0.005 \approx 2\sigma$). The solver stops at step-norm 10^{-5} or 20 iterations. Normals are estimated via PCA over a 0.15 m neighborhood (minimum of 5 points), and the keyframe is reset whenever the inlier ratio drops below 20%. Two keyframe-splitting criteria are considered: a translation (0.50 m–3.00 m) and an inlier-ratio threshold (40%–90%).

b) Evaluation metrics: Drift is assessed with relative metrics computed over $\Delta d = 10$ m trajectory segments [11] using the evo framework [23] on the estimated and ground-truth TUM [24] trajectories. For each segment relative error ε_s with translational $\mathbf{t}_{\varepsilon_s}$ and rotational θ_{ε_s} parts over N_s segments, the Relative Translational Error (RTE) and Relative Rotational Error (RRE) are computed as in (12), where RTE [25] expresses drift as a percentage of the segment length.

$$\text{RTE} = \frac{100\%}{N_s} \sum_{s=1}^{N_s} \frac{\|\mathbf{t}_{\varepsilon_s}\|_2}{\Delta d}, \quad \text{RRE} = \frac{1}{N_s} \sum_{s=1}^{N_s} \frac{|\theta_{\varepsilon_s}|}{\Delta d} \quad (12)$$

Both RTE and RRE are reported as the mean over all segments [11]. These relative metrics are preferred over the absolute trajectory error to isolate the local consistency of the laser-odometry front-end independently of global alignment. Table IV reports the odometric drift results.

c) Laser odometry drift: As in the synthetic benchmark, $\mathcal{D}_{p2pl}(\mathcal{M}_0)$ and $\text{ICP}_{\mathcal{D}, p2pl}$ produce numerically equivalent trajectories (34 and 38 of the 48 configurations differing by less than 0.05% RTE and $0.005^\circ \text{ m}^{-1}$ RRE, respectively), reaffirming the theoretical equivalence under real tracking conditions. At the $\mathcal{M}_0$ level, the point-to-plane formulations outperform their point-to-point counterparts across the majority of configurations, especially on RTE, consistent with the synthetic results and with Rusinkiewicz and Levoy [12]. The exceptions concentrate on Slippage and at the 80%–90% inlier thresholds, where point-to-plane degrades for both the distance map and ICP variants on all sequences at 80% and on Nav A Diff, Nav A Omni, and Slippage at 90%, indicating that the degradation stems from the error formulation rather than the correspondence mechanism. At 90%, the point-to-plane advantage persists only on the Loop sequence, whose doorway transitions keep point-to-point residuals limited by the sampling discrepancy.

The precomputed point-to-point formulation $\mathcal{D}_{p2p}(\mathcal{M}_1)$ proposed in this work outperforms both $\mathcal{D}_{p2p}(\mathcal{M}_0)$ and $\text{ICP}_{\mathcal{D}, p2p}$ across most configurations (e.g., lower RTE in 39 and lower RRE in 35 of the 48 cases against $\text{ICP}_{\mathcal{D}, p2p}$), and even reaches competitive drift against the point-to-plane formulations on Nav A Diff, Nav A Omni, and Slippage, with best figures of $0.95\%/0.078^\circ \text{ m}^{-1}$, $0.81\%/0.065^\circ \text{ m}^{-1}$, and $0.55\%/0.056^\circ \text{ m}^{-1}$, respectively. This result is attributed to the finite-difference gradient acting as an implicit point-to-plane constraint within planar regions while regularizing sensor noise, without requiring explicit normal estimation. The residual gap persists on the Loop sequence (2.14% against 1.37% for $\text{ICP}_{\mathcal{D}, p2pl}$), where gradient directions deviate from surface normals near the doorway discontinuities. The

TABLE IV
REAL-WORLD LASER ODOMETRY DRIFT ON THE IILABS 3D [11] DATASET USING PROPOSED DISTANCE MAP-BASED FORMULATIONS
AND DISTANCE MAP-GUIDED ICP BASELINES [1], [2] FOR THE 2D POINT CLOUD REGISTRATION PROBLEM

Method	Splitting Criteria	Threshold	Nav A Diff		Nav A Omni		Loop		Slippage	
			RTE (%)	RRE ($^{\circ}$ m $^{-1}$)	RTE (%)	RRE ($^{\circ}$ m $^{-1}$)	RTE (%)	RRE ($^{\circ}$ m $^{-1}$)	RTE (%)	RRE ($^{\circ}$ m $^{-1}$)
$\mathcal{D}_{p2p}(\mathcal{M}_0)$	Translation (m)	0.50 m	2.91	0.182	2.64	0.134	5.80	0.310	1.28	0.059
		1.00 m	2.05	0.161	1.94	0.108	4.35	0.236	0.96	0.074
		1.50 m	1.76	0.146	1.63	0.098	3.03	0.183	0.78	0.056
		2.00 m	1.51	0.127	1.50	0.102	3.13	0.202	0.97	0.049
		2.50 m	1.35	0.126	1.23	0.088	3.86	0.199	0.68	0.033
		3.00 m	<u>1.47</u>	0.120	<u>1.45</u>	0.121	3.91	0.198	0.92	0.060
	Inliers Ratio (%)	40%	1.12	0.098	1.09	0.090	2.54	0.158	0.71	0.034
		50%	1.35	0.122	1.31	0.089	3.00	0.144	0.64	0.052
		60%	1.58	0.129	1.39	0.090	3.12	0.175	0.63	0.052
		70%	1.83	0.141	1.66	0.098	4.37	0.217	0.95	0.082
		80%	2.53	0.174	2.17	0.127	5.34	0.256	1.24	0.064
		90%	5.34	0.292	3.98	0.187	9.65	0.399	2.80	0.093
$\mathcal{D}_{p2p}(\mathcal{M}_1)$	Translation (m)	0.50 m	1.62	0.109	1.38	0.080	2.87	0.140	1.47	0.088
		1.00 m	1.12	0.096	1.01	0.077	3.09	0.160	0.89	0.088
		1.50 m	1.11	0.098	0.98	0.080	2.81	0.154	1.04	0.142
		2.00 m	1.06	0.091	0.92	0.079	2.82	0.155	0.55	0.056
		2.50 m	<u>1.12</u>	<u>0.091</u>	0.91	0.080	3.07	0.166	0.85	0.086
		3.00 m	1.09	0.091	1.36	0.123	3.03	0.156	0.88	0.069
	Inliers Ratio (%)	40%	1.01	0.090	0.89	0.087	2.14	0.134	0.66	0.076
		50%	0.95	0.078	0.81	0.065	2.33	0.123	0.64	0.070
		60%	1.10	0.089	0.89	0.080	2.80	0.143	0.85	0.087
		70%	1.15	0.094	0.97	0.063	3.01	0.155	0.81	0.080
		80%	1.37	0.112	1.15	0.075	3.18	0.160	0.89	0.083
		90%	1.92	0.131	1.61	0.067	3.99	0.133	2.05	0.144
$\mathcal{D}_{p2pl}(\mathcal{M}_0)$	Translation (m)	0.50 m	1.60	0.145	1.35	0.103	4.13	0.275	1.48	0.149
		1.00 m	1.36	0.115	1.30	0.106	3.75	0.239	1.01	0.091
		1.50 m	1.25	0.126	1.19	0.096	2.88	0.210	0.77	0.077
		2.00 m	1.16	0.115	0.88	0.064	2.39	0.163	0.71	0.097
		2.50 m	1.08	0.103	1.05	0.105	2.91	0.189	0.66	0.081
		3.00 m	<u>1.67</u>	0.114	1.25	0.108	3.59	0.140	0.75	0.090
	Inliers Ratio (%)	40%	0.92	0.094	1.09	0.104	1.41	0.107	0.84	0.073
		50%	1.06	0.104	0.84	0.086	2.05	0.139	0.72	0.084
		60%	1.10	0.111	0.85	0.084	2.77	0.156	0.68	0.056
		70%	1.24	0.124	1.06	0.090	3.69	0.212	0.69	0.056
		80%	3.38	0.154	5.40	0.233	7.10	0.253	5.01	0.084
		90%	8.02	0.216	9.48	0.214	8.24	0.139	7.06	0.251
$\mathcal{D}_{p2pl}(\mathcal{M}_2)$	Translation (m)	0.50 m	1.34	0.128	1.07	0.097	4.43	0.279	0.90	0.084
		1.00 m	1.30	0.120	1.11	0.109	3.93	0.246	0.92	0.097
		1.50 m	1.24	0.121	1.12	0.102	3.83	0.227	0.78	0.051
		2.00 m	<u>1.13</u>	0.103	1.05	0.103	3.21	0.188	1.00	0.053
		2.50 m	1.13	0.106	0.98	0.093	3.50	0.189	0.81	0.060
		3.00 m	1.22	0.123	1.24	0.118	2.26	0.156	0.97	0.112
	Inliers Ratio (%)	40%	1.03	0.087	1.00	0.085	2.09	0.124	0.92	0.098
		50%	1.02	0.090	0.92	0.087	3.18	0.148	1.06	0.133
		60%	1.24	0.108	1.11	0.104	3.24	0.168	0.98	0.050
		70%	1.37	0.129	1.11	0.106	3.87	0.225	1.16	0.062
		80%	2.60	0.174	2.62	0.216	7.61	0.250	7.99	0.240
		90%	14.02	0.630	12.10	0.577	11.33	0.611	16.03	0.472
ICP $_{\mathcal{D}, p2p}$ [1]	Translation (m)	0.50 m	3.03	0.180	2.56	0.133	5.63	0.297	1.20	0.070
		1.00 m	2.01	0.154	1.76	0.117	4.46	0.235	0.57	0.067
		1.50 m	1.56	0.133	1.50	0.107	2.97	0.180	0.70	0.052
		2.00 m	1.40	0.123	1.19	0.100	<u>2.56</u>	<u>0.180</u>	0.67	0.056
		2.50 m	1.24	0.110	1.13	0.082	3.55	0.181	0.67	0.045
		3.00 m	1.33	0.113	1.59	0.095	3.60	0.186	0.67	0.038
	Inliers Ratio (%)	40%	1.10	0.099	1.07	0.092	3.36	0.192	0.56	0.055
		50%	1.32	0.116	1.16	0.082	2.95	0.158	0.72	0.065
		60%	1.48	0.133	1.21	0.100	3.15	0.169	0.64	0.048
		70%	1.80	0.147	1.60	0.110	4.32	0.209	0.88	0.070
		80%	2.54	0.169	2.14	0.111	5.24	0.247	0.83	0.057
		90%	5.14	0.298	4.01	0.177	10.01	0.403	2.63	0.099
ICP $_{\mathcal{D}, p2pl}$ [2]	Translation (m)	0.50 m	1.61	0.146	1.42	0.111	4.14	0.275	1.48	0.149
		1.00 m	1.37	0.114	1.28	0.103	3.71	0.238	1.06	0.095
		1.50 m	1.25	0.122	1.19	0.097	2.82	0.207	0.77	0.077
		2.00 m	1.13	0.115	0.91	0.070	2.45	0.164	0.71	0.097
		2.50 m	1.09	0.095	<u>1.02</u>	0.105	3.02	0.187	0.66	0.081
		3.00 m	<u>1.68</u>	0.112	1.22	0.103	3.55	0.140	0.75	0.090
	Inliers Ratio (%)	40%	0.88	0.082	1.06	0.106	1.37	0.106	0.78	0.073
		50%	1.01	0.097	0.83	0.093	2.07	0.140	0.72	0.084
		60%	1.08	0.109	0.82	0.075	2.71	0.154	0.82	0.070
		70%	1.24	0.123	0.93	0.090	3.91	0.207	0.69	0.056
		80%	3.34	0.155	5.39	0.231	7.32	0.257	4.92	0.081
		90%	7.98	0.214	9.45	0.217	8.26	0.136	6.97	0.251

Legend: distance map representation level ($\mathcal{M}_l$); Relative Translational Error (RTE); Relative Rotational Error (RRE); RTE (%) and RRE ($^{\circ}$ m $^{-1}$) reported as mean over 10m segments. Bold entries indicate lowest drift within 0.05 % RTE and 0.005 $^{\circ}$ m $^{-1}$ RRE. Underlined entries indicate lowest drift within method-splitting setups.

precomputed point-to-plane variant $\mathcal{D}_{p2pl}(\mathcal{M}_2)$ is competitive under the translation-based criterion but yields the highest RTE of all formulations at the 90% inlier threshold, mirroring its noise sensitivity observed in the synthetic benchmark.

Finally, the keyframe-splitting criterion has an effect on drift. The translation-based criterion performs best between 2.0 m and 2.5 m in 19 of the 24 method–sequence pairs. The inlier-ratio criterion performs best at 40%–60%, degrading at 80% and above, where frequent keyframe splits accumulate registration and numerical errors. The 40% inlier threshold yields lower or equal RTE than the best translation threshold in 15 of the 24 pairs, suggesting that a data-driven policy adapts to varying tracking conditions better than fixed distances.

VI. CONCLUSIONS

This paper proposes a 2D point cloud registration framework based on unsigned distance maps using $O(1)$ grid lookups over a precomputed distance field. We derive point-to-point and point-to-plane residuals on the $SE(2)$ manifold. On the real-world IILABS 3D dataset [11], the precomputed point-to-point variant $\mathcal{D}_{p2p}(\mathcal{M}_1)$ reduces drift compared to analytical point-to-point baselines across most configurations, even achieving drift levels competitive with point-to-plane formulations. We attribute this performance to the finite-difference gradient, which acts as an implicit point-to-plane constraint in planar surfaces while smoothing sensor noise. Uncertainty consistency analysis indicates that analytical $\mathcal{M}_0$ point-to-plane methods are the least overconfident, making them preferable for uncertainty-aware applications such as SLAM. Nevertheless, $\mathcal{D}_{p2p}(\mathcal{M}_1)$ yields less overconfident point-to-point information matrices than its analytical counterparts in degenerate scenes. Future work will integrate our framework into a full SLAM pipeline.

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